

Solutions to
Introduction to Modern Cryptography (Third
Edition)
by Katz & Lindell

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2025

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Chapter 1

Introduction and Classical Cryptography

1.1 Decrypt the ciphertext provided at the end of the section on mono-alphabetic substitution ciphers.

To decrypt it, let's play with C++ and create some classes and helper methods. The source code is available at [KatzCryptoBookSolutions repository on Zer0Leak's GitHub](https://github.com/Zer0Leak/KatzCryptoBookSolutions)

Step 01: Finding the word *the*

The word *the* is the most common word in english (see <https://norvig.com/ngrams/>). That's why we go for it.

Launch the program with `auto step = kStep01_1`; This prints:

VFP	-	4
FPF	-	4
QVF	-	4

Table 1.1: Possible words with three letters in cipherText that appears at least 4 times

Notice that only *VFP* or *QVF* may be decrypted to *the*.

Launch the program with `auto step = kStep01_2`; This executes:

```
1 AlphabetTools::attackMonoAlphaCipher(_englishAlphabet
    , cipherText, knownDict, true);
```

And it will print a side-by-side comparison between cipher text and general english letter probabilities.

F (0.152)	<->	e (0.127)
Q (0.107)	<->	t (0.091)
W (0.086)	<->	a (0.082)
G (0.078)	<->	o (0.075)
L (0.070)	<->	i (0.070)
O (0.066)	<->	n (0.067)
V (0.061)	<->	s (0.063)
H (0.057)	<->	h (0.061)
B (0.049)	<->	r (0.060)
P (0.041)	<->	vd (0.043)
J (0.037)	<->	l (0.040)
I (0.037)	<->	u (0.028)
Z (0.029)	<->	c (0.028)
R (0.029)	<->	w (0.024)
M (0.016)	<->	m (0.024)
E (0.016)	<->	f (0.022)
Y (0.012)	<->	y (0.020)
K (0.012)	<->	g (0.020)
C (0.012)	<->	p (0.019)
A (0.012)	<->	b (0.015)
S (0.008)	<->	v (0.010)
D (0.008)	<->	k (0.008)
X (0.004)	<->	x (0.002)
U (0.000)	<->	j (0.002)
T (0.000)	<->	z (0.001)
N (0.000)	<->	q (0.001)

Table 1.2: Side-by-side cipheText and English letters probabilities

In additon, it prints the sum square differece of probabilties of from *VFP* and *QVF* to *the*.

QVF	->	0.000849
VFP	->	0.016486

Table 1.3: Sum square distance probabilities mapping to 'the'

Visually looking the table of sorted probabilities, *QVF* seems to be better translated into *the*, in addition, the sum square distance above shows it is much better, *i.e.* the sum square distance is much smaller.

Lauch the program with *auto step = kStep01_3*; This will set the translation table and print the cipherText decryption:

Q	t
V	h
F	e

Table 1.4: Tranlation table known until this step.

```
JGRMQOYGHMVB JW RWQFPWHGFFDQGFPFZRKBEEBJIZQQOCIBZKLFAFGQVFZFWWEO
GWOPFGFHWOLPHRLLOLFD MF GQWBLWBWQOLKFWBYLBYLFSFLJGRMQBOLWJVFPFW
QVHQWFFPQOQVFPQOCFPOGFWFJIGFQVHLHLROQVFGWJVFPFOLFHGQVQVFILEOGQ
ILHQFQGIQVVOSFAFGBWQVHQWIJWVJVFPFWHGFIWIHZZRQGBABHZQOCGFHX

__t__h__te__ee_t_e_e__tt__e_e_the_e__
__e_e__e_e_t__t__e__e_e__t__he_e_
th_t_ee_t_the_t__e__e_e__eth__the__he_e_e_ththe__t
__tet__thh__e_e__th_t__h__he_e__e__t__t__e__
```

Step 02: Finding the word *that*.

The word *that* is one of the most common words in english ,and we will benefit from *t* and *h* we have already found.

Lauch the program with *auto step = kStep02_1*;

For this we will replace all characters in cipher text that correspond to letters *t*, *h*, and *e*.

This will print all the words that with 4 letters that repeats at least 2 times.

WJhe	-	3
JheP	-	3
hePe	-	3
WthH	-	2
JGRM	-	2
hHtW	-	2
ePeW	-	2
WHGe	-	2
GRMt	-	2
Othe	-	2
eAeG	-	2
thHt	-	2
ePtO	-	2

Table 1.5: Words with four letters that repeats at least 2 times.

Notice that *thHt* is the only one that can match *that*. Then we assume that *H* translates into *a*.

Looking Table 1.2 show that *H* probability is 0.057 and *a* probability is 0.082, these are not to close, but we decided to take the risk.

One more mathematical approach could be to see check square distance of *QVFH* to *THEA* we just found, against a different choice in first step, leading to *VFP?* to *THEA* in this step. (? is the character we would find here if we take a VFP in the first step).

Also, instead of print Table 1.2 in step *kStep02_1*, we could narrow down to print only words that matches *tha.t*, but we wanted to print the table to see if something interesting could grab our attention.

Launch the program with *auto step = kStep02_2*; This will increment the translation table and print the cipherText decryption:

Q	t
V	h
F	e
H	a

Table 1.6: Translation table known until this step.

JGRMQOYGHMVB	JWRWQFPWHGFFDQGF	PfZrKBEEBJIZQQOCIBZKLFAFGQVFZFWWEO
GWOPFGFWOLPHLR	LOLFDMFGQWBLWBWQOLKFWBYLBLYLFSFLJGRMQBOLWJVFPFW	
QVHQWFFPQOQVFP	QOCFPOGFWFJIGFQVHLHLROQVFGWJVFPFOLFHGQVQVFILEOGQ	


```

ILHQFQGIQVVOFAGBWQVHQWIJVVJVPFWHGFIWIHZZRQGBABHZQOCGFHX

____t__a_h____te__a_ee_t_e_e_____tt_____e_e_the_e____
____e_ea____a____e_e_t_____t__e_____e_e____t____he_e_
that_ee_t_the_t__e____e_e____etha_a____the____he_e_ea_ththe_____t
__atet__thh__e_e__that__h__he_e_a_e__a__t____a_t__ea_

```

Now, what bring me attention was the sequence *e__ea_ththe*, actually *a_th* or *ea_th*, because there is no word with *thth*, let's find words that matches *^a.th\$* or *.*ea.th\$*. Also notice that in case of *.*ea.th\$*, the two chars before *ea*, can't be any in *thea*.

index:	count	-	word
1332:	61059905	-	earth
4722:	14325239	-	wealth
13096:	3287162	-	auth
16550:	2207395	-	breadth
34718:	623206	-	anth
43011:	430700	-	unearth
47176:	367476	-	dearth
58228:	256042	-	acth
59047:	249807	-	ealth
65661:	208067	-	arth
84092:	134308	-	ayth
185308:	34516	-	xearth
193319:	32100	-	aith
219593:	26605	-	aeth
249908:	21239	-	alth

Table 1.7: Words matching regex (*^a.th\$* / *.*ea.th\$*) with additional constraints

We could map *eaGth* into *earth*, *i.e.* map *G* with 0.078 into *r* with 0.060. And *earth* is the most common between them.

Or, we could map *LeaGth* into *wealth*, *i.e.* map *G* with 0.078 into *l* with 0.040, and map *L* with 0.070 into *w* with 0.024. *wealth* is still very common. But this leads to a bad probability match. One could try to map with it and see if translation shows something interesting.

Also, we could map *aGth* into *auth*, i.e. map *G* with 0.078 into *u* with 0.028. The word *auth* not that common but is related to the subject of the book. But the probabilities are not good either.

I will try mapping *eaGth* into *earth*. And see if something interesting follows.

Launch the program with *auto step = kStep03_2*; This will increment the translation table and print the cipherText decryption:

Q	t
V	h
F	e
H	a
G	r

Table 1.8: Translation table known until this step.

```
JGRMQOYGHMVBJWRWQFPWHGFFDQGFPFZRKBEEBJIZQQOCIBZKLFAFGQVFZFWWEO
GWOPFGFHWOLPHRLOLFDMFGQWBLWBWQOLKFWBYLBLYLFSFLJGRMQBOLWJVFPFW
QVHQWFFPQOQVFPQOCFPOGFWFJIGFQVHLHLROQVFGWJVFPFOLFHGQVQVFILEOGQ
ILHQFQGIQVVSFAFGBWQVHQWIJVVJVFPFWHGFIWIHZZRQGBABHZQOCGFHX

_r__t__ra_h_____te__aree_tre_e_____tt_____e_erthe_e____
r____erea____a_____e_ert_____t___e_____e_e_r__t_____he_e_
that_ee_t_the_t__e__re_e__retha_a____ther__he_e__earththe____rt
__atetr_thh__e_er__that___h__he_e_are___a___tr___a_t__rea_
```

From now on I will follow similar approach, trying to find words, looking letter's probabilities, checking if translation seems weird or good. Launch the program with the next steps to see the output of each step.

If something letter really pulls our attention and we know for sure its translation, then fine, we add it to translation table, but if we are not sure, we try to match a letters that appears more and helps more on next step to find words or weird combinations.

The final result is:

```
JGRMQOYGHMVBJWRWQFPWHGFFDQGFPFZRKBEEBJIZQQOCIBZKLFAFGQVFZFWWEO
GWOPFGFHWOLPHRLOLFDMFGQWBLWBWQOLKFWBYLBLYLFSFLJGRMQBOLWJVFPFW
QVHQWFFPQOQVFPQOCFPOGFWFJIGFQVHLHLROQVFGWJVFPFOLFHGQVQVFILEOGQ
ILHQFQGIQVVSFAFGBWQVHQWIJVVJVFPFWHGFIWIHZZRQGBABHZQOCGFHX
```

cryptographicsystemsareextremelydifficulttobuildneverthelessfor
 somereasonmanynonexpertsinsistondesigningnewencryptionschemes
 thatseemtothemtobemoresecurethananyotherschemeonearththeunfort
 unatetruthhoweveristhatsuchschemesareusuallytrivialtobreak

1.2 Provide a formal definition of the **Gen**, **Enc**, and **Dec** algorithms for the mono-alphabetic substitution cipher.

Equating the English alphabet with the set $\{0, \dots, 25\}$ (so $a = 0$, $b = 1$, etc.), the message space $\mathcal{M}$ is then any finite sequence of integers from this set, *i.e.* message $m = m_1 \cdots m_\ell$ (where $m_i \in \{0, \dots, 25\}$).

Let the encryption key $K = (k_0, k_1, \dots, k_{25})$ be an ordered sequence of 26 elements such that:

1. $k_i \in \{0, 1, \dots, 25\}$ for all i .
2. $k_i \neq k_j$ for all $i \neq j$ (*i.e.*, all elements are unique).

Gen is function that creates a key k , which is any permutation of $\{0, \dots, 25\}$ with $26!$ possibilities.

Let **Enc** be a function that encrypts a message $\mathcal{M} = (m_1, m_2, \dots, m_\ell)$ with a given key k . The encryption is defined as:

$$\mathbf{Enc}_k(m_1, m_2, \dots, m_\ell) = (c_1, c_2, \dots, c_\ell) \quad \text{where} \quad c_i = k(m_i) = k_{m_i}$$

Let **Dec** be a function that decrypts a chiphered message $\mathcal{C} = (c_1, c_2, \dots, c_\ell)$ with a given key k .

Let g be a inverse of k function, *i.e.* $g(k(m_i)) = m_i$ and $k(g(c_i)) = c_i \forall i$

I.e., given that j is the index in k where c_i is found, therefore $j \in \{0, \dots, 25\}$, we have $g(c_i) = j$.

The decryption is defined as:

$$\mathbf{Dec}_k(c_1, c_2, \dots, c_\ell) = (m_1, m_2, \dots, m_\ell) \quad \text{where} \quad m_i = g(c_i)$$

1.3 Provide a formal definition of the **Gen**, **Enc**, and **Dec** algorithms for the Vigenère cipher. (Note: there are several plausible choices for **Gen**; choose one.)

Equating the English alphabet with the set $A = \{0, \dots, 25\}$ (so $a = 0$, $b = 1$, etc.), the message space $\mathcal{M}$ is then any finite sequence of integers from this set, *i.e.* message $m = m_1 \cdots m_\ell$ (where $m_i \in \{0, \dots, 25\}$).

Let the encryption key $k = (k_0, k_1, \dots, k_n)$ be an ordered sequence of n finite natural integer elements such that $k_i \in A \forall i$ and n is a finite integer number. So, for each n there are 26^n different keys.

Gen is a function that generates a key k , which is an ordered selection with repetition (or a permutation with repetition) of elements from the set $\{0, \dots, 25\}$ in n positions. And is defined as:

$$\mathbf{Gen}_n = (k_0, k_1, \dots, k_{n-1}) \quad \text{where} \quad k_i \in \{0, \dots, 25\}$$

Let **Enc** be a function that encrypts a message $\mathcal{M} = (m_0, m_1, \dots, m_{\ell-1})$ with a given key k of length n . The encryption is defined as:

$$\mathbf{Enc}_k(m_0, m_1, \dots, m_{\ell-1}) = (c_0, c_1, \dots, c_{\ell-1}) \quad \text{where} \quad c_i = k(m, i), \quad \text{and} \quad k(m, i) \text{ is defined as :}$$

$$\mathbf{h}(m, i) = c_i = [(m_i + k_j) \bmod 26] \quad \text{where} \quad j = i \bmod n$$

Let **Dec** be a function that decrypts a ciphered message $\mathcal{C} = (c_0, c_1, \dots, c_{\ell-1})$ with a given key k of length n . The decryption is defined as:

$$\mathbf{Dec}_k(c_0, c_1, \dots, c_{\ell-1}) = (m_0, m_1, \dots, m_{\ell-1}) \quad \text{where} \quad m_i = g(c, i), \quad \text{and} \quad g(c, i) \text{ is defined as :}$$

$$\mathbf{g}(c, i) = m_i = [(c_i - k_j) \bmod 26] \quad \text{where} \quad j = i \bmod n$$

1.4 Say you are given a ciphertext that corresponds to English-language text that was encrypted using either the shift cipher or the Vigenère cipher with period greater than 1. How could you tell which was the case?

Shift cipher is a special case of Vigenère where the key length is 1.

If the key length is 1, and q_i is the probability of occurrence of i -th letter in cipher text. This means that, if the key is j , $q_{(i+j) \bmod 26} \approx p_i$, where p_i is the probability of occurrence of i -th letter in english texts. I.e. probabilities are close to the same, with shifted j index in q and p . Therefore, it is ShiftCipher, check the following:

$$\sum_{i=0}^{25} q_i^2 \approx \sum_{i=0}^{25} p_i^2 \approx 0.065$$

If you want to double try to find key length using *index of coincidence method* as described in text book and implemented at:

vigenereattack.cpp: _findKeyLength on **chap_01** solutions at [Zer0Leak's solutions of KatzCryptoBookSolutions on GitHub](#).

1.5 Implement the attacks described in this chapter for the shift cipher and the Vigenère cipher.

See **chap_01** solutions at [Zer0Leak's solutions of KatzCryptoBookSolutions on GitHub](#).

1.6 The shift and Vigenère ciphers can also be defined on the 256-character alphabet consisting of all possible bytes (8-bit strings), and using XOR instead of modular addition.

(a) Provide a formal definition of both schemes in this case.

Let's provide the formal definition of Vigenère cipher. The formal definition of Shift cipher is a special case of it where the encryption key is $K = (k_0)$ i.e. the encryption key has length 1.

Let the *Sign* alphabet to be the set $\{0x00, \dots, 0xFF\}$ (where each symbol is an unsigned 8-bits integer), the message space $\mathcal{M}$ is then any finite sequence from this set, i.e. message $m = m_1 \cdots m_\ell$ (where $m_i \in \text{Sign}$).

Let the encryption key $K = (k_0, k_1, \dots, k_w)$ be a finite ordered sequence of w elements such that $k_i \in \text{Sign}$ for all i .

Gen is a function that creates a key k with a finite length w , which is any permutation of $\{0x00, \dots, 0xFF\}$ with 255^w possibilities.

Let **Enc** be a function that encrypts a message $\mathcal{M} = (m_1, m_2, \dots, m_\ell)$ with a given key k . The encryption is defined as:

$$\mathbf{Enc}_k(m_1, m_2, \dots, m_\ell) = (c_1, c_2, \dots, c_\ell) \quad \text{where} \quad c_i = m_i \text{ XOR } k_{(i \bmod w)}$$

Let **Dec** be a function that decrypts a ciphered message $\mathcal{C} = (c_1, c_2, \dots, c_\ell)$ with a given key k . The decryption is defined as:

$$\mathbf{Dec}_k(c_1, c_2, \dots, c_\ell) = (m_1, m_2, \dots, m_\ell) \quad \text{where} \quad m_i = c_i \text{ XOR } k_{(i \bmod w)}$$

(b) Discuss how the attacks we have shown in this chapter can be modified to break these schemes.

The attacks discussed in this chapter, and implemented in exercise 1.5 can be modified as follow:

The **Gen** function now uses alphabet $\{0x00, \dots, 0xFF\}$ instead of $\{0, \dots, 25\}$.

The **Enc** and **Dec** functions now, for each symbol uses $f_i = s_i \text{ XOR } k_{(i \bmod w)}$ instead of $f_i = (s_i \pm k_{(i \bmod w)}) \bmod 26$

1.7 The index of coincidence method relies on a known value for the sum of the squares of plaintext-letter frequencies (cf. Equation (1.1)). Why would it not work using the $\sum_i p_i$ itself?

First of all, $\sum_i p_i$ is always 1 (*i.e.* 100%). So, it is useless.

But also, using square sum, uneven distribution will lead to higher values of sumation, while uniform will result in smaller values. So, using squared sum will help in find the statistic distribution that is closer to english distribution.

1.8 Show that the shift, substitution, and Vigenère ciphers are all trivial to break using a chosen-plaintext attack. How much chosen plaintext is needed to recover the key for each of the ciphers?

In **chosen-plaintext attack** there is a cipherd text of your knowledge you want to discover its plain-text pair. You don't know the key though. But you can choose any plain-text of your choice to cipher using the same unknown key. Thus you can have any pair of text/cipher text of you choice generated by the same key that was used to cipher the text under attack.

For the Shift cipher, just use the plain text "a" wich is integer 0, the key is $k = c_0$. Then use the just discovered key to break the cipher text under attack.

For the Vigenère cipher, just use the plain text "aaa..." wich are integers 000..., for each symbol i , the key symbol at index i is $k_i = c_i$. The length of the plain-text you need it at most the lenght of the cipher text you want to break. Then, from k , if it has a repeating sequence, extract a simpler key from it. Then use the just discovered key to break the cipher text under attack.

1.9 Assume an attacker knows that a user's password is either **bcda** or **bedg**. Say the user encrypts his password using the shift cipher, and the attacker sees the resulting ciphertext. Show how the attacker can determine the user's password, or explain why this is not possible.

It is trivial to discover.

bcda is 1230 and bedg is 1436. So, using Shift cipher will just shift these by $k \bmod 26$. But the difference between them must remain the same.

Consider the cipher text to be $c_0c_1c_2c_3$.

If key is **bcda**:

$$\begin{aligned}(c_1 - c_0) \bmod 26 &= 1 \\(c_2 - c_0) \bmod 26 &= 2 \\(c_3 - c_0) \bmod 26 &= 25\end{aligned}$$

If key is **bedg**:

$$\begin{aligned}(c_1 - c_0) \bmod 26 &= 3 \\(c_2 - c_0) \bmod 26 &= 2 \\(c_3 - c_0) \bmod 26 &= 5\end{aligned}$$

It is still possible to break by comparing c_1 and c_0 , or c_3 and c_0 . You can compare indexes if their difference is not the same, *i.e.* comparing c_2 and c_0 in this case is useless

1.10 Repeat the previous exercise for the Vigenère cipher using period 2, using period 3, and using period 4.

(a) Period 2

If key is **bcda**:

$$\begin{aligned}(c_2 - c_0) \bmod 26 &= 2 \\(c_3 - c_1) \bmod 26 &= 24\end{aligned}$$

If key is **bedg**:

$$\begin{aligned}(c_2 - c_0) \bmod 26 &= 2 \\(c_3 - c_1) \bmod 26 &= 2\end{aligned}$$

So, it is still possible to break by comparing c_3 and c_1

(a) Period 3

If key is **bcda**:

$$(c_3 - c_0) \bmod 26 = 25$$

If key is **bedg**:

$$(c_3 - c_0) \bmod 26 = 5$$

So, it is still possible to break by comparing c_3 and c_0

(a) Period 4

It is impossible to discover.

1.11 The attack on the Vigenère cipher has two steps: (a) find the key length by identifying with $S_\tau \approx 0.065$ (cf. Equation (1.3)) and (b) for each character of the key, find j maximizing I_j (cf. Equation (1.2)), using $\{p_i\}$ corresponding to English text. What happens in each case if the underlying plaintext is in a language other than English?

If the language has a different sum of square of probabilities, and we don't know that, we will notice that for multiple τ the value will be closely the same. And then we can figure out τ and the language's sum of square of probabilities. Considering it is a latin alphabet of 26 letters, from 'a' to 'z', for wrong values of τ , the sum will tend towards 0.038, i.e. $\sum_{i=0}^{25} \left(\frac{1}{26}\right)^2$

For the second part, equation 1.2, it will result in wrong decryption producing garbage.

Chapter 2

Perfectly Secret Encryption

2.1 Prove that, by redefining the key space, we may assume that the key-generation algorithm **Gen** chooses a uniform key from the key space, without changing $\Pr[C = c | M = m]$ for any m, c .

Motivation: It's easier to think mathematically with a **Gen** that generates keys at a uniform distribution.

So, let's use a new key ω that is uniformly distributed over a new key space Ω . And replace the **Gen**() = **k** by the uniform **GenU**() = **w**. We also define **Map**(**w**) = **k** in a way that, the distribution of $K = \text{Map}(\text{Gen}(W))$ is the same as the original.

As an example, let's say $K = \{0, 1\}$ with $\Pr[K = 0] = 0.75$ and $\Pr[K = 1] = 0.25$. We then choose $\mathbf{W} = \{00, 01, 10, 11\}$ with equal probability (*uniform distribution*), and make $\text{Map}(w) = 1$ if $w = 11$, otherwise $\text{Map}(w) = 0$.

With the new key space we have:

$$\Pr[C = c \mid M = m] = \sum_{w \in \mathcal{W}} \Pr[\text{GenU}() = w] \cdot \Pr[\text{Enc}_{\text{map}(w)}(m) = c]$$

Also notice that for each k in the original scheme can be produced by one or more ω in the new scheme. That is:

$$\Pr[\text{Gen}() = k] = \sum_{\omega: \text{Map}(\omega) = k} \Pr[\text{GenU}() = \omega] = \sum_{\omega} \Pr[\text{GenU}() = \omega] \cdot \Pr[\text{Enc}_{\text{Map}(\omega)}(m) = c]$$

Let's prove that from the original key space.

$$\Pr[C = c \mid M = m] = \sum_{k \in \mathcal{K}} \Pr[\text{Gen}() = k] \cdot \Pr[\text{Enc}_k(m) = c]$$

$$= \sum_{k \in \mathcal{K}} \left(\sum_{w: \text{map}(w)=k} \Pr[\text{Gen}U() = w] \right) \cdot \Pr[\text{Enc}_k(m) = c]$$

Flatting the summation we get the new schema:

$$= \sum_{w \in \mathcal{W}} \Pr[\text{Gen}U() = w] \cdot \Pr[\text{Enc}_{\text{map}(w)}(m) = c]$$

2.2 Prove that, by redefining the key space as well as the encryption algorithm, we may assume that encryption is deterministic without changing $\Pr[C = c | M = m]$ for any m, c .

Motivation: It's easier to think mathematically with a deterministic Enc instead of randomized.

If $\text{Enc}_k(m)$ is randomized, it generates $c \in C_{m,k} = \{c_0, c_1, \dots, c_l\}$ chipered messages. If we compute then randomized part ω , separately, we can rewrite Enc as $\text{Enc}_k(m, \omega)$.

We can now move out the internal random part and make it as part of the Key. So, the Key is now $k\omega$. Then:

$\text{Gen}() = k$ becomes $\text{Gen}'() = k\omega$

$\text{Enc}_k(m)$ becomes $\text{Enc}'_{k\omega}(m) := \text{Enc}_k(m, \omega)$

$\text{Dec}_k(c)$ becomes $\text{Dec}'_{k\omega}(c)$ that is equal to the original $\text{Dec}_k(c)$, because Dec is deterministic and doesn't depend on ω .

From original schema:

$$\Pr[C = c | M = m] = \sum_k \Pr[k] \sum_{\omega} \Pr[\omega] \cdot \Pr[\text{Enc}_k(m, \omega) = c]$$

In new schema:

$$\Pr[C = c | M = m] = \sum_{k\omega} \Pr[k\omega] \cdot \Pr[\text{Enc}'_{k\omega}(m) = c]$$

$$\Pr[C = c | M = m] = \sum_{k\omega} \Pr[k\omega] \cdot \Pr[\text{Enc}_k(m, \omega) = c]$$

Now, since $\Pr[k\omega] = \Pr[k] \cdot \Pr[\omega]$ (*i.e.* they are independent), the expression becomes:

$$\Pr[C = c \mid M = m] = \sum_k \sum_{\omega} \Pr[k] \Pr[\omega] \cdot \Pr[\text{Enc}_k(m\omega) = c]$$

$$\Pr[C = c \mid M = m] = \sum_k \Pr[k] \sum_{\omega} \Pr[\omega] \cdot \Pr[\text{Enc}_k(m\omega) = c]$$

Which is the same as the original.

2.3 Prove or refute: An encryption scheme with message space $\mathcal{M}$ is perfectly secret if and only if for every probability distribution on $\mathcal{M}$ and every $c_0, c_1 \in \mathcal{C}$ we have $\Pr[C = c_0] = \Pr[C = c_1]$.

This is false as we can see from encryption scheme below.

$$\begin{aligned} \mathcal{M} &= m_0 = \{0, 1\} \\ \mathcal{K} &= k_0 k_1 r \in \{0, 1\}^3 = \{000, 001, 010, \dots, 110, 111\} \\ \mathcal{C} &= c_0 c_1 \in \{0, 1\}^2 = \{00, 01, 10, 11\} \\ \text{Enc}_k(m) &= c_0 c_1 \text{ where } c_0 = m \oplus k_0 \oplus r \text{ and } c_1 = r \oplus k_1 \\ \text{Dec}_k(m) &= m_0 = c_0 \oplus k_0 \oplus r \text{ where } r = c_1 \oplus k_1 \end{aligned}$$

Then, let's assume the $\Pr[M = 0] = 0.5, \Pr[k_0 = 0] = 0.5, \Pr[k_1 = 0.25] = 0.5, \Pr[r = 0] = 0.25$.

m	k_0	k_1	r	$m \oplus k_0 \oplus r$	$k_1 \oplus r$	C	$\Pr(C)$
0	0	0	0	0	0	00	0.015625
0	0	0	1	1	1	11	0.046875
0	0	1	0	0	1	01	0.046875
0	0	1	1	1	0	10	0.140625
0	1	0	0	1	0	10	0.015625
0	1	0	1	0	1	01	0.046875
0	1	1	0	1	1	11	0.046875
0	1	1	1	0	0	00	0.140625
1	0	0	0	1	0	10	0.015625
1	0	0	1	0	1	01	0.046875
1	0	1	0	1	1	11	0.046875
1	0	1	1	0	0	00	0.140625
1	1	0	0	0	0	00	0.015625
1	1	0	1	1	1	11	0.046875
1	1	1	0	0	1	01	0.046875
1	1	1	1	1	0	10	0.140625
sum							1

Notice it is still perfectly secret

$\Pr[C \mid M = 0]$		$\Pr[C \mid M = 1]$	
C	Prob	C	Prob
00	0.15625	00	0.15625
01	0.09375	01	0.09375
10	0.15625	10	0.15625
11	0.09375	11	0.09375
sum	0.5	sum	0.5
sum of sums		1	

But not all $c \in C$ has same probability.

$\Pr[C = c_0] \neq \Pr[C = c_1]$	
C	Prob
00	0.3125
01	0.1875
10	0.3125
11	0.1875
sum	1

2.4 Prove or refute: For every perfectly secret encryption scheme it holds that for every distribution on the message space $\mathcal{M}$, every $m, m' \in \mathcal{M}$ and every $c \in \mathcal{C}$:

$$\Pr[\mathcal{M} = m \mid \mathcal{C} = c] = \Pr[\mathcal{M} = m' \mid \mathcal{C} = c]$$

From $\Pr[\mathcal{M} = m \mid \mathcal{C} = c] = \Pr[\mathcal{M} = m]$ (Lemma 2.3 - definition of perfect secret)

$$\frac{\Pr[\mathcal{M} = m \mid \mathcal{C} = c]}{\Pr[\mathcal{M} = m]} = 1$$

Bayes' Theorem gives:

$$\Pr[M = m \mid C = c] = \frac{\Pr[C = c \mid M = m] \cdot \Pr[M = m]}{\Pr[C = c]}$$

Therefore:

$$\frac{\Pr[M = m \mid C = c]}{\Pr[M = m]} = \frac{\Pr[C = c \mid M = m]}{\Pr[C = c]} = 1$$

and:

$$\frac{\Pr[M = m' \mid C = c]}{\Pr[M = m']} = \frac{\Pr[C = c \mid M = m']}{\Pr[C = c]} = 1$$

then:

$$\Pr[C = c \mid M = m] = \Pr[C = c \mid M = m']$$

2.5 Prove that in Definition 2.6 we may assume $\mathcal{A}$ is deterministic without loss of generality.

From the experiment. The encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$, and the randomness of $k \leftarrow \text{Gen}$ and the choice of what message will be encrypted, *i.e.*, the choice of bit b , is not under the control of adversary $\mathcal{A}$.

The adversary $\mathcal{A}$, can only choose two messages $m_0, m_1 \in \mathcal{M} \mid m_0 \neq m_1$, and the output bit b' .

So, let's fix the experiment, *i.e.* the part adversary $\mathcal{A}$ has not control of.

Then, consider $\mathcal{A}$ is random. Since there are a finite set of combinations of $m_0, m_1 \in \mathcal{M} \mid m_0 \neq m_1$ and b' , there is one of these.

For a random $\mathcal{A}$, we have:

$$\Pr[\text{Priv}_{A,\Pi}^{\text{eav}} = 1] = \frac{1}{2}$$

Now, let's take the random part out of $\mathcal{A}$, with $r \leftarrow \mathcal{R}$, and rewrite a deterministic one as $\mathcal{A}'(r)$. Now we have:

$$\Pr[\text{Priv}_{A,\Pi}^{\text{eav}} = 1] = \sum_{r \in \mathcal{R}} \Pr[\text{Priv}_{A'(r),\Pi}^{\text{eav}} = 1] \cdot \Pr[R = r] = \mathbb{E}_r[\Pr[\text{Priv}_{A'(r),\Pi}^{\text{eav}} = 1]]$$

I.e. it is the average or expected value of $\Pr[\text{Priv}_{A'(r),\Pi}^{\text{eav}} = 1]$ with $r \in \mathcal{R}$.

So, there is a r' where:

$$\Pr[\text{Priv}_{A'(r'),\Pi}^{\text{eav}} = 1] \geq \mathbb{E}_r[\Pr[\text{Priv}_{A'(r),\Pi}^{\text{eav}} = 1]] = \Pr[\text{Priv}_{A,\Pi}^{\text{eav}} = 1] = \frac{1}{2}$$

But, we can't allow $\Pr[\text{Priv}_{A'(r'),\Pi}^{\text{eav}} = 1] > \frac{1}{2}$ for any r , because, if we do so, then there would be some randomness that would choose just those good r s and then $\Pr[\text{Priv}_{A,\Pi}^{\text{eav}} = 1] > \frac{1}{2}$. And we can't allow any r so that $\Pr[\text{Priv}_{A'(r'),\Pi}^{\text{eav}} = 1] < \frac{1}{2}$, because it would make $\mathbb{E}_r[\Pr[\text{Priv}_{A'(r),\Pi}^{\text{eav}} = 1]] < \frac{1}{2}$.

Then, finally:

$$\Pr[\text{Priv}_{A'(r'),\Pi}^{\text{eav}} = 1] = \Pr[\text{Priv}_{A,\Pi}^{\text{eav}} = 1] = \frac{1}{2}$$

2.6 Prove Lemma 2.7.

First let's prove that if it is perfect secret, then $\Pr[\text{Priv}_{A,\Pi}^{\text{eav}} = 1]$ is $\frac{1}{2}$.

$$\begin{aligned}\Pr[\text{Priv}_{A,\Pi}^{\text{eav}} = 1] &= \Pr[b' = b] \\ &= \Pr[b' = 0 \mid M = m_0] \Pr[M = m_0] + \Pr[b' = 1 \mid M = m_1] \Pr[M = m_1]\end{aligned}$$

$\mathcal{A}$ is deterministic, then, let's assume that if $c \in C_0$ it outputs $b' = 0$ if $c \in C_1$ it outputs $b' = 1$.

$$\begin{aligned}&= \Pr[c \in C_0 \mid M = m_0] \Pr[b = 0] + \Pr[c \in C_1 \mid M = m_1] \Pr[b = 1] \\ &= \Pr[c \in C_0 \mid M = m_0] \cdot \frac{1}{2} + \Pr[c \in C_1 \mid M = m_1] \cdot \frac{1}{2} \\ &= \sum_{c \in C_0} \Pr[C = c \mid M = m_0] \cdot \frac{1}{2} + \sum_{c \in C_1} \Pr[C = c \mid M = m_1] \cdot \frac{1}{2}\end{aligned}$$

If it is perfectly secret, then (from Definition 2.3 and Bayes' theorem): $\Pr[C = c \mid M = m] = \Pr[C = c]$. Therefore:

$$\begin{aligned}&= \sum_{c \in C_0} \Pr[C = c] \cdot \frac{1}{2} + \sum_{c \in C_1} \Pr[C = c] \cdot \frac{1}{2} \\ &= \frac{1}{2} (\sum_{c \in C_0} \Pr[C = c] + \sum_{c \in C_1} \Pr[C = c])\end{aligned}$$

Also notice that $C_0 \cup C_1 = C$ and $C_0 \cap C_1 = \emptyset$. Then:

$$= \frac{1}{2} \cdot \sum_{c \in C} \Pr[C = c] = \frac{1}{2}$$

To prove the opposite direction, *i.e.*, if $\Pr[\text{Priv}_{A,\Pi}^{\text{eav}} = 1] = \frac{1}{2}$, then it is perfectly secret. We will do contrapositive, *i.e.*, if **not** perfectly secret than it is **not** perfectly indistinguishable.

If **not** perfectly secret, then, from definition, there must be a m' and c' where:

$$\Pr[M = m' \mid C = c'] \neq \Pr[M = m']$$

Also, notice that:

$$\begin{aligned}\sum_{m \in \mathcal{M}} \Pr[M = m] &= 1 \\ &= \sum_{m \in \mathcal{M} \setminus \{m'\}} \Pr[M = m] + \Pr[M = m']\end{aligned}$$

And

$$\begin{aligned}\sum_{m \in \mathcal{M}} \Pr[M = m \mid C = c'] &= 1 \\ &= \sum_{m \in \mathcal{M} \setminus \{m'\}} \Pr[M = m \mid C = c'] + \Pr[M = m' \mid C = c']\end{aligned}$$

Notice that we **can't** assume that for all $m \in \mathcal{M} \setminus \{m'\} \Rightarrow \Pr[M = m \mid C = c'] \neq \Pr[M = m]$

Because, if we do so, in equations above, it would force that $\Pr[M = m' \mid C = c'] = \Pr[M = m']$, and it is the opposite of our assumption.

So, there must be at least another m'' that $\Pr[M = m' \mid C = c'] \neq \Pr[M = m'']$

Now let's define the deterministic adversary $\mathcal{A}$ (See exercise 2.5), so that

$$m_0 = m', m_1 = m''$$

Then, now let's consider our message space as $\mathcal{M} = \{m_0, m_1\}$.

$$\Pr[M = m \mid C = c'] \neq \Pr[M = m] = \frac{1}{2}$$

$$\Pr[M = m \mid C = c'] = \frac{\Pr[C=c' \mid M=m] \cdot \Pr[M=m]}{\Pr[C=c']} \quad (\text{Bayes' Theorem})$$

$$\Pr[C = c' \mid M = m] = \frac{\Pr[M=m \mid C=c']}{\Pr[M=m]} \cdot \Pr[C = c']$$

$$\Rightarrow \Pr[C = c' \mid M = m] \neq \Pr[C = c']$$

The adversary knows $\mathcal{A}$ that $\Pr[M = m_0 \mid C = c'] \neq \Pr[M = m_1 \mid C = c']$, and then, one is greater than $\frac{1}{2}$, and the other is smaller. Let's suppose $\Pr[M = m_0 \mid C = c'] > \frac{1}{2} > \Pr[M = m_1 \mid C = c']$.

Let's also define adversary $\mathcal{A}$ to output $b' = 0$ if $c = c'$, otherwise it outputs $b' \leftarrow \{0, 1\}$. So, that:

$$\begin{aligned} \Pr[\text{Priv}_{A,\Pi}^{\text{eav}} = 1] &= \Pr[b' = b] \\ &= \Pr[b' = 0 \mid M = m_0] \Pr[M = m_0] + \Pr[b' = 1 \mid M = m_1] \Pr[M = m_1] \\ &= \frac{1}{2} \cdot (\Pr[b' = 0 \mid M = m_0] + \Pr[b' = 1 \mid M = m_1]) \end{aligned}$$

let's break down $\Pr[b' = 0 \mid M = m_0]$

$$\begin{aligned} &= \Pr[C = c' \mid M = m_0] + \frac{1}{2} \Pr[C \neq c' \mid M = m_0] \\ &= \Pr[C = c' \mid M = m_0] + \frac{1}{2} (1 - \Pr[C = c' \mid M = m_0]) \\ &= \frac{1}{2} + \frac{1}{2} \cdot \Pr[C = c' \mid M = m_0] \end{aligned}$$

let's break down $\Pr[b' = 1 \mid M = m_1]$

$$= \frac{1}{2} \Pr[C \neq c' \mid M = m_1]$$

So:

$$\begin{aligned} \Pr[\text{Priv}_{A,\Pi}^{\text{eav}} = 1] &= \frac{1}{2} \left(\frac{1}{2} + \frac{1}{2} \cdot \Pr[C = c' \mid M = m_0] + \frac{1}{2} \Pr[C \neq c' \mid M = m_1] \right) \\ &= \frac{1}{4} (1 + \Pr[C = c' \mid M = m_0] + \Pr[C \neq c' \mid M = m_1]) \\ &\neq \frac{1}{4} (1 + \Pr[C = c' \mid M = m_1] + \Pr[C \neq c' \mid M = m_1]) = \frac{1}{4} (1 + 1) = \frac{1}{2} \end{aligned}$$

Therefore:

$$\Pr[\text{Priv}_{A,\Pi}^{\text{eav}} = 1] \neq \frac{1}{2}$$

2.7 What is the ciphertext that results when the plaintext 0x012345 (written in hex) is encrypted using the one-time pad with key 0xFFEEDD?

Enc is defined as $Enc_k(m) = m \oplus k$, so $c = 0x012345 \oplus 0xFFEEDD = 0xFECD98$

2.8 For each of the following encryption schemes, state whether the scheme is perfectly secret. Justify your answer in each case.

- (a) The message space is $\mathcal{M} = \{0, \dots, 4\}$, and Gen chooses a uniform key from the key space $\mathcal{K} = \{0, \dots, 5\}$. $Enc_k(m)$ returns $[m + k \bmod 5]$, and $Dec_k(c)$ returns $[c - k \bmod 5]$.
- (b) The message space is $\mathcal{M} = \{m \in \{0, 1\}^\ell \mid \text{the last bit of } m \text{ is } 0\}$. Gen chooses a uniform key from $\{0, 1\}^{\ell-1}$. $Enc_k(m)$ returns ciphertext $m \oplus (k \parallel 0)$, and $Dec_k(c)$ returns $c \oplus (k \parallel 0)$.

- (a) It is not perfectly secret.

Suppose $c = 0$ (notice we can generalize it of any value of c from 0 to 4). It can happen in the following cases:

$$M = 0 \wedge k = 5 \Rightarrow \Pr[M = 0 \wedge k = 5 \mid C = 0] = 1/6$$

$$M = 1 \wedge k = 4 \Rightarrow \Pr[M = 1 \wedge k = 4 \mid C = 0] = 1/6$$

$$M = 2 \wedge k = 3 \Rightarrow \Pr[M = 2 \wedge k = 3 \mid C = 0] = 1/6$$

$$M = 3 \wedge k = 2 \Rightarrow \Pr[M = 3 \wedge k = 2 \mid C = 0] = 1/6$$

$$M = 4 \wedge k = 1 \Rightarrow \Pr[M = 4 \wedge k = 1 \mid C = 0] = 1/6$$

$$M = 0 \wedge k = 0 \Rightarrow \Pr[M = 0 \wedge k = 0 \mid C = 0] = 1/6$$

$$\text{Therefore } \Pr[M = 0 \mid C = 0] = 2/6 \neq 1/5 = \Pr[M = 0]$$

- (b) Is perfectly secret.

It doesn't make sense to have $\ell = 1$, because we would only have one message, 0, and not available key, making no sense encrypt it. So, it is essentially a One-Time Pad encryption.

The last bit of message plays no role at all and doesn't change anything. So the proof is exactly the same found in Theorem 2.10, by just replacing all occurrences of $2^{-\ell}$ by $2^{-(\ell-1)}$

2.9 In each of the following schemes, $\text{Enc}_k(m) = [m + k \bmod 3]$. State in each case whether the scheme is perfectly secret, and justify your answers.

- (a) The message space is $\mathcal{M} = \{0, 1\}$, and **Gen** chooses a uniform key from the key space $\mathcal{K} = \{0, 1\}$.
- (b) The message space is $\mathcal{M} = \{0, 1, 2\}$, and **Gen** chooses a uniform key from the key space $\mathcal{K} = \{0, 1, 2\}$.
- (c) The message space is $\mathcal{M} = \{0, 1\}$, and **Gen** chooses a uniform key from the key space $\mathcal{K} = \{0, 1, 2\}$.
- (a) **Not** perfectly secret.

$k \backslash m$	0	1
0	0	1
1	1	2

e.g. $\Pr[M = 0 \mid C = 0] = 1 \neq \Pr[M = 0]$ or $\Pr[M = 1 \mid C = 2] = 1 \neq \Pr[M = 1]$

- (b) Perfectly secret.

$k \backslash m$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$$\Pr[M = m \mid C = c] = \Pr[M = m]$$

We could also use Shannon's theorem because $|\mathcal{M}| = |\mathcal{K}| = |\mathcal{C}|$.

(1) holds, k is uniform

(2) holds, noticed there is only one value of c per column, and all columns have all possible c . I.e. there is exactly one c per column and none is missing. Notice there is no requirement for one c per line in theorem, but it is required to make scheme correct, otherwise Dec_k is no deterministic.

- (c) Perfectly secret.

$k \backslash m$	0	1
0	0	1
1	1	2
2	2	0

$$\Pr[M = m \mid C = c] = \Pr[M = m]$$

2.10 The following questions concern the message space $\mathcal{M} = \{0, 1\}^{\leq \ell}$, the set of all nonempty binary strings of length at most ℓ .

- (a) Consider the encryption scheme in which **Gen** chooses a uniform key from $\mathcal{K} = \{0, 1\}^\ell$, and $\text{Enc}_k(m)$ outputs $k_{|m|} \oplus m$, where k_t denotes the first t bits of k . Show that this scheme is not perfectly secret for message space $\mathcal{M}$.
- (b) Design a perfectly secret encryption scheme for message space $\mathcal{M}$.
- (a) It is easy to show. Consider $\ell = 2$

$k \backslash m$	0	1	00	01	10	11
00	0	1	00	01	10	11
01	0	1	01	00	11	10
10	1	0	10	11	00	01
11	1	0	11	10	01	00

Notice that $\Pr[M = 0 \mid C = 0] = 2/4 = 1/2 \neq 1/6 = \Pr[M = 0]$

- (b) As follow:

If $\ell \in \{2^k \mid k \in \mathbb{N}\}$, *i.e.* ℓ is a power of 2. Then we can use the following approach.

- (i) Let $\mathbf{s}_1$ to be a binary string with $\lceil \log_2 \ell \rceil$ bits
- (ii) Encode $\mathbf{s}_1$ with unsigned binary representation of $m-1$, *i.e.* $\mathbf{s}_1 = \text{bin}(|m| - 1)$, we do -1 because there is no $|m| = 0$, and we don't want to waste $000\dots 0$ with nothing.
- (iii) Cypher $\mathbf{s}_1$ with $\mathbf{s}_1' = k_{\lceil \log_2 \ell \rceil} \oplus \mathbf{s}_1$
- (iv) Let r be $r \leftarrow \{0, 1\}^{\ell - |m|}$, *i.e.* uniform random bit string with $\ell - |m|$ bits.

$$\text{Enc}'_k(m) = \mathbf{s}_1' \parallel k_{|m|} \oplus m \parallel r$$

Notice that all $c \in C$ now has exactly $\lceil \log_2 \ell \rceil + \ell$ bits.

$$\text{Dec}'_k(c) \Rightarrow$$

$$\mathbf{s}_1 = \mathbf{s}_1' \oplus k_{\lceil \log_2 \ell \rceil}, \text{ notice } \mathbf{s}_1 = |m| - 1$$

$$c' = c_{[\lceil \log_2 \ell \rceil .. \lceil \log_2 \ell \rceil + \mathbf{s}_1 + 1]}, \text{ i.e. substring starting at } \lceil \log_2 \ell \rceil, \text{ and length } \mathbf{s}_1 + 1 = |m|$$

$$\text{Dec}'_k(c) = k_{|s_1+1|} \oplus c' = m$$

If $\ell \notin \{2^k \mid k \in \mathbb{N}\}$, *i.e.* ℓ is **not** a power of 2. Then we can use the following approach.

Add $\lceil \log_2 \ell \rceil$ bits to k , *i.e.* $\mathcal{K} = \{0, 1\}^{\ell + \lceil \log_2 \ell \rceil}$

Now encrypt/decrypt $\mathbf{s}_1$ with the last $\lceil \log_2 \ell \rceil$ bits of k . And do similar to scheme above.

This is required because if ℓ is not power of 2, then $|m| = \mathbf{s}_1 + 1 = \mathbf{s}_1' \oplus k_{\lceil \log_2 \ell \rceil} + 1$ can't be greater than ℓ , and so, some $k_{\lceil \log_2 \ell \rceil}$ are not allowed, giving a hint of $k_{\lceil \log_2 \ell \rceil}$ bits, and then a hint on message because some $m_{\lceil \log_2 \ell \rceil} = k_{\lceil \log_2 \ell \rceil} \oplus c_{\lceil \log_2 \ell \rceil}$ are not allowed too. Using then ending k bits that are only used to cipher message size won't hint the message or message size itself.